COMP5424 Extended Reality

# Prototype Design and Technical Note

Phase 2 – Group 7

Scenario D: Campus Emergency Preparedness

VR Stair-Evacuation and Lift-Safety Practice

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|---|---|
| **Virtual client** | Harbourview University Safety and Resilience Office |
| **Submission date** | [To confirm: date] |
| **Repository** | https://github.com/itemgroup/COMP5424_CampusEmergency_XR |
| **Source baseline** | feat/final-vr-demo; updated source in PR #22 |
| **Submitted build** | [To confirm: ZIP filename and matching build identifier] |
| **Target-PC evidence** | [To confirm: PQ604A/B/C platform and launch record] |


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*A supervised educational demonstration; not official emergency guidance, an approved campus evacuation plan, or certification.*

# 1 Prototype Scope and Requirements Traceability

## 1.1 Purpose and implemented scope

Phase 1 proposed sign interpretation and recovery when an initial evacuation route becomes unavailable [1]. Phase 2 prioritises a runnable, testable XR journey rather than visual polish [2]. The current source implements **stair-evacuation and lift-safety practice** with an optional, authored route-closure and recovery exercise.

**Implemented journey:** Enter System → welcome and task briefing → stair route and fire alert → marked landings → 2F practice endpoint and persistent completion card. Optional reroute mode first presents the original route, then closes it at the 3F decision corner and requires a signposted same-floor alternative before rejoining the stairs. The diagram and controller tutorial remain optional. A separate G-floor courtyard assembly interaction is additional content, not the 2F route's completion condition.

## 1.2 Requirement-to-feature mapping

Status describes software scope, not demonstrated learning effectiveness. Evidence keys refer to Appendix A and the source baseline [3].

| ID | Requirement / implementation | Status / evidence |
|---|---|---|
| R1 | Complete evacuation practice journey: briefing, stairs, endpoint and optional recovery. | Implemented authored practice; learning effect not validated. S1–S3; V1, V3. |
| R2 | Interpret signs and environmental cues: floor/lift signs, diagram and task card. | Implemented cues; independent interpretation not established. S1–S3; V2. |
| R3 | Recover when an initial route becomes unavailable. | Optional 3F closure, enforced alternative and stair rejoin; authored, not dynamic routing. V3. |
| R4 | Explain why an action or route is appropriate. | Lift warning and safety review explain stair/lift choice and closure cues; learning effect unmeasured. S2–S4; V3. |
| R5 | Calm, inclusive, low-risk practice. | Partial: text feedback, optional tutorial, teleport and moderate audio settings; comfort needs user evidence. V4. |
| R6 | Pause, exit and restart safely. | Implemented controls; full device verification remains open. S2; V5. |

## 1.3 Changes from Phase 1

The route-change requirement is implemented as a bounded, optional exercise using one fixed closure, signs and an authored corridor, not as dynamic routing. Reusing the VR template and imported building assets enabled integration of movement, signs and feedback. The environment includes multiple floors, exceeding the proposed small scene. Campus-derived geometry is a visual setting, not a validated real evacuation map. The prototype can be evaluated for task completion and cue usability, but current formative evidence cannot establish learning or transfer. AI co-navigation, multiplayer, certification and graphic emergency effects remain outside scope.

# 2 Interaction and Prototype Design

## 2.1 Core interaction and route decisions

The welcome NPC introduces the demonstration. Continue reveals the task list, evacuation diagram and optional five-step controller practice. Users then read corridor signs and enter the marked stair route. Arrival at its starting waypoint starts the exercise audio; ordered checkpoints update the route task state. Position-based triggers support both walking and teleport arrival.

Four floors contain eight lift signs. Nearby panels appear within 2.5 m horizontally and 1.25 m vertically of a sign. During the fire exercise, entering this area can announce that lifts must not be used; activating a panel adds warning text and an error cue. Panels sit 0.5 m above and 0.5 m along world $+Z$ from each sign to reduce surface intersection.

**Decision and rationale:** Fixed markers make arrival feedback repeatable and avoid dependence on NPC placement. The optional reroute exercise closes the initial 3F route, blocks invalid teleport destinations and cross-barrier shortcuts, then signposts an opposite same-floor corridor back to the existing stairs. It is an authored scenario, not dynamic route planning. Outside this exercise ordinary exploration is unchanged. The diagram's arrows and automatic lift announcement reveal directions, so they confound claims of unaided choice.

## 2.2 Locomotion, controls, and feedback

**Input:** The left stick moves; the right stick turns and its upward teleport input aims at a valid target. Release confirms teleport, the right grip cancels, and the right trigger interacts with supported objects/UI. Teleport reduces reliance on continuous travel but comfort is not guaranteed. The tutorial still describes 45-degree snap turning; this wording must be checked against the delivered turn configuration.

**UI:** Tutorial/navigation panels are world-fixed after opening. Nearer/Farther physically move a panel in 0.2 m steps (normal range 0.65–2.2 m); Review reopens the guide at the current viewpoint. Thus the panel does not chase head movement, although users can leave it behind. The completion card persists after arrival at the endpoint or End NPC, independently of Continue. It offers a safety review, a no-op Continue and a Back action that reloads the scene to clear the previous attempt and open the classroom welcome page. The review describes stairs, lift avoidance, noticing a closure and the simulated endpoint; it is not evidence of retained learning.

**Pause/reset:** Pause sets the simulation time scale to zero and opens a view-following menu; Resume restores it. Restart reloads the active scene; Exit quits the player. This is not proof that every XR input is disabled.

## 2.3 Calmness, accessibility, and journey evidence

English text, symbols, high-contrast panels and adjustable panel distance provide visual support alongside audio. Low-intensity cues avoid deliberate flashes, smoke or graphic harm. Multilingual access, seated reach, silent-mode comprehension and individual comfort are not yet established.

# 3 Technical Implementation and Functional Verification

## 3.1 Architecture and major technical decisions

**Configuration:** Unity 6000.3.25f1; URP 17.3.0; XR Interaction Toolkit 3.5.1; Input System 1.20.0; OpenXR 1.17.1; Android IL2CPP/ARM64 for Quest 3 [3]. Lab-PC compatibility is not established.

**Control:** ControllerGuidePanel tracks start, stair entry, two landings and completion using actual teleport-anchor positions. An independent 2.5 m check requests the End NPC card once without advancing tasks. Optional RouteChangeExercise adds a global XRI selection filter, closure barrier and alternative signs only after opt-in. NpcGuideDialogue maintains cards; MainMenuController restores campus roots on entry and supports a clean classroom reload; CampusAudioDirector coordinates audio; LiftSafetyAudio handles signs.

**Trade-offs:** One scene preserves the XR rig; authored markers and offline speech reduce integration/network risk. Optional recovery is bounded to one authored closure and corridor. The ordinary endpoint card may appear without ordered completion; reroute practice additionally requires recovery and the original stair checkpoints in order.

## 3.2 Executable and repository reproducibility

Development used Windows, an RTX 5060 Laptop GPU and Quest 3 Android deployment. The latest previously recorded test_2 Build And Run succeeded on 17 October at 00:37; it predates the current reroute/reset source update and does not verify this version. **PQ604A/B/C extraction and launch are unverified.**

**Handoff:** [To confirm: ZIP, launch path, companion files and lab-PC record]. Deploy the current APK to Quest, launch it and select Enter System. The cover identifies repository and source commit; access and ZIP/source correspondence require confirmation. Required member accounts must still be added to the root README.

## 3.3 Functional verification summary

C# compilation and **144 isolated Unity regression checks passed** for the current source, including XR teleport filtering, barrier collision, ordered recovery and two clean scene reloads. These are automated regressions, not participant trials or full-campus acceptance (Appendix A).

| Key | Executed or required check | Result / evidence |
|---|---|---|
| V1 | Endpoint before Continue; persistent card; task isolation. | Pass; ordinary mode can acknowledge arrival before ordered completion. |
| V2 | Diagram, Continue, panel movement and limits. | Pass; exact labels in evidence file. |
| V3 | Optional closure, barrier, valid recovery and ordered stairs. | Pass in isolated scene; authored scenario, not participant evidence. |
| V4 | Lift proximity; muted journey. | Proximity passes; muted journey not recorded. |
| V5 | Pause, clean classroom reset and exit. | Reset passes in isolated fixture; full device checks not recorded. |
| V6 | Extracted build on lab PC. | Not run / not evidenced. |

# 4 Limitations and Evaluation Priorities

## 4.1 Operational definitions: recovery and assistance

Phase 1 feedback requested explicit criteria for successful recovery and acceptable assistance. The following is a **provisional Phase 3 protocol**, not a measured result or client-approved safety standard. It becomes applicable only after an unavailable-route event is implemented.

**Successful recovery:** After the closure cue appears, the participant recognises the closure through observable behaviour or explanation, chooses the signposted alternative, and reaches the fictional endpoint without bypass. Pilot a 120-second active-time window from cue presentation to endpoint entry, excluding pauses and technical interruptions, then freeze the threshold before the main study. Record recognition, route selection and completion separately.

**Assistance:** H0 = no route-recovery hints; H1 = at most one non-directional prompt (e.g., "Look again at the signs"); H2 = repeated prompts, explicit directions or facilitator-led navigation. Log wording, source and time, including system prompts. Standard control briefing is separate from route help. The current arrow diagram/automatic directions must be recorded as directional assistance, not silently counted as H0.

**Judgement:** Report completed recoveries as H0 independent, H1 acceptably assisted, or H2 beyond the proposed allowance. Report timeouts, incomplete attempts, technical interruptions and comfort withdrawals separately, with denominators. Always provide safety/comfort help regardless of classification.

## 4.2 Evaluation priorities and evidence

**E1 – Cue interpretation (R2):** Observe missed signs, hesitation and control errors; ask which cue informed each choice. **E2 – Recovery/rationale (R1, R3, R4):** The closure, alternative and safety review are now implemented. Record event times and assistance; ask for rationale before the review and record post-review explanations separately. **E3 – Comfort/inclusion (R5, R6):** Observe reading/reach difficulty, pause/exit use and discomfort; participation is voluntary.

**Study status:** Group members provided iterative hands-on feedback, summarised in Appendix A. This is retrospective formative convenience testing, not an independent study. The number present at each iteration, individual attribution, exact test dates, device/build and standardised outcomes were not logged. GitHub commit dates bound the development period but are not treated as test timestamps. No success rate, time saving or learning gain is claimed. Future evaluation should record anonymous task outcomes, prompts and optional comments with consent; do not retain names, account IDs or headset recordings by default.

## 4.3 Known limitations and next priorities

**1. Establish recovery for unfamiliar users (R1/R3/R4):** The closure, alternative and review now exist, but there is no independent evidence that users notice the closure, choose the alternative or understand the rationale. **2. Establish delivery reliability (R6):** Verify a matching ZIP on the required lab platform, reset/exit and final controls. **3. Pilot clarity and assistance (R2/R5):** Check mute comprehension, turning text, fixed-panel reach and sign alignment on-device. Teammate familiarity, a single layout and absent transfer evidence limit conclusions.

## References

[1] Group 7. *Requirements and Prototype Proposal: Scenario D — Campus Emergency Preparedness*. COMP5424 Extended Reality, 20 September 2026. Repository copy: GroupProjectDocuments/Group7-GroupComponent.pdf.

[2] COMP5424 Extended Reality. *Phase 2 assignment instructions: Purpose, Minimum Prototype Expectations and Submission Components*. Course instructions supplied by the team; publication date/version not provided. Includes the four-page technical-note and target-platform requirements.

[3] Group 7. *Campus Emergency XR prototype*, source baseline 416ca44 and follow-on final-source update in PR #22, 17 October 2026. https://github.com/itemgroup/COMP5424_CampusEmergency_XR. Implementation evidence: Packages/manifest.json, ProjectSettings/ProjectVersion.txt, ProjectSettings/ProjectSettings.asset, Docs/SoundImplementation.md, and Tools/Sound/Validation/SoundVerification.cs.

# A Evidence Locations and Submission Checks

## A.1 Source locators

Paths are relative to the repository root. PR #22 carries the final-source update beyond 416ca44. No new APK or submission ZIP was built for this report update; verify the submitted package against its source revision.

| Key | Locator and interpretation |
|---|---|
| S1 | Assets/Scenes/SampleScene.unity: Start/End NPCs, DownStair waypoints, LiftSign and startup menu. |
| S2 | Assets/Scripts/UI/ControllerGuidePanel.cs: LateUpdate, AdvanceRoute, TryShowEndDialogue, AdjustPanelDistance, pause/restart/exit handlers. |
| S3 | Assets/Scripts/Building/NpcGuideDialogue.cs: ShowIntroduction, ShowTaskGuide, ShowCompletionDialogue. |
| S4 | Assets/Scripts/Audio/CampusAudioDirector.cs: IsNearLiftSign, WarnAgainstLift, UpdateEvacuation; Assets/Scripts/Audio/LiftSafetyAudio.cs. |

## A.2 Executed regression evidence

The accompanying functional-verification.txt contains the saved PASS record and 144 named checks from an isolated, no-graphics Unity project. The persistent test definitions are in Tools/Sound/Validation/SoundVerification.cs; execute using Tools/Sound/Verify-Sound.ps1 with the Unity editor path. Representative exact labels:

- V1: “Completion card remains visible beyond its former seven-second timeout”; “Completion before Continue plays or queues the configured completion voice”.
- V2: “Guide Nearer physically moves its panel 0.2 meters without changing height or orientation”; “HUD Farther clamps to 2.2 meters”.
- V3: “The registered selection filter rejects a real stair anchor before recovery”; “The original five stair checkpoints still complete in order after reroute recovery”.
- V4: “NPC proximity cannot show the lift warning even when explicitly requested”.
- V5: “Back skips Enter System and its music, opening the fresh classroom welcome instead”; “Reload clears closure, task checkmarks, endpoint success and prior completion UI”. The suite uses an isolated fixture and does not verify the entire campus or Quest build.

## A.3 Retrospective group-member formative test log

Group members report hands-on iterations covering the full route, optional reroute, lift-safety prompts, controller/UI use and the Back reset. The issue notes below are consolidated from development feedback, not a contemporaneous test protocol. Individual attribution, per-session headcount, exact test dates, headset/build identifiers and per-person task outcomes were not retained. The GitHub development history spans 12–17 October 2026; this is a development window, not a claim that each test occurred on those commit dates.

| Task / observation | Feedback recorded | Response and evidence |
|---|---|---|
| Controls and UI | Navigation/tutorial panels were reported to follow the player or distance controls appeared ineffective. | Panels are world-fixed; Nearer/Farther move them in bounded 0.2 m steps. Automated checks cover transform stability and distance limits. |
| Lift safety and completion | Lift warning appeared near the NPC rather than the lift signs; completion feedback was also reported missing or misplaced. | Lift prompts use the eight actual LiftSign markers. Completion is triggered by the final anchor or End NPC and remains visible. Regression checks cover marker proximity and endpoint behaviour. |
| Safety review | The Review safety button was obscured by the NPC head. | The shared completion/review card layout was raised; panel-bound checks pass in the isolated regression suite. |
| Optional reroute | The first route sign appeared mirrored and invalid-teleport notices recurred frequently. | Sign orientation was corrected; teleport validation now checks destination and barrier crossing, and rate-limits feedback per held gesture. Regression checks cover four viewing directions and repeated selection. |
| Back reset | Returning to spawn left traces of the previous attempt. | Back now reloads the scene and opens the classroom welcome directly. The final reset-flow check was reported as passed on 17 October 2026; the device and build used for this acceptance were not recorded. Automated tests also verify two consecutive clean reloads. |

### A.4 Items the team must complete before submission

- Fill the cover date, ZIP identifier and platform launch evidence. Confirm repository visibility or teaching-team access. The root README now lists each member's name, student ID and GitHub account; the report cover intentionally lists only names and student IDs.
- Supply the separate weekly activity record with actual responsibilities, decisions, completed work and unresolved issues; do not infer individual contributions from this report.
- Preserve source licences/credits and verify redistribution rights for imported models, images and menu music.
- Keep this retrospective feedback separate from a future independent evaluation; log tester count, device/build, task outcomes and assistance at the time of testing. Recompile this PDF after replacing blue fields; retain four body pages at 12 pt.